

Jaswanth KS

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PROFESSIONAL SUMMARY

Software Engineer with around **3** years of experience building **backend services**, **data pipelines**, and cloud-based applications using **Python** and **TypeScript**. Experienced in designing data ingestion systems, **REST APIs**, and data enrichment workflows across structured datasets. Provisioned and managed **AWS** infrastructure using **Terraform**, and worked with containerized deployments using **Docker**, with exposure to **Kubernetes**. Recently worked on prototype workflow using **LLM APIs** and retrieval-based approaches to explore document search, summarization, and enrichment across structured and document datasets. Completed an **MSc in Data Science and AI** (Merit), with a focus on **machine learning**, information retrieval, and **Applied AI** engineering. Interested in applying **GenAI** and retrieval-based systems to automate workflows and build intelligent agent-assist tooling at scale.

WORK EXPERIENCE

Software Engineer

Nov 2021 – Aug 2023

Growz Software Solutions

Bengaluru, India

- Designed and built **data ingestion** and enrichment pipelines integrating structured business and financial datasets into **AWS**-based data platforms, improving data accessibility and query performance by **30%**.
- Developed **microservices** and **REST APIs** using **Python** and **TypeScript**, to integrate external APIs and ML services, and automated data tagging workflows.
- Contributed to **ML pipeline** tooling and **RAG**-based workflows using **LangChain** and **LLM APIs** prototyping document search, summarisation, and **AI-assisted features**; collaborated with data science teams on model evaluation and output quality.
- Implemented **CI/CD** pipelines using **Git** and **Jenkins** and automated **AWS** infrastructure provisioning using **Terraform**, reducing deployment cycle time by **40%**.
- Collaborated with product and data teams to improve data modelling and support downstream analytics and **ML** workflows.

Software Engineer Intern

Sep 2020 – Oct 2021

Techborn IT Solutions

Bengaluru, India

- Developed backend services and REST APIs using **Python** to support predictive modelling workflows, analytics dashboards, and operational reporting tools
- Assisted in developing and testing classification models using **TensorFlow**, and **Keras**.
- Built preprocessing pipelines using **Pandas** and **NumPy** for dataset organisation tasks.

SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, SQL, T-SQL
- **Backend Development & APIs:** Fast API, Flask, Django, REST APIs, Microservices
- **AI / ML & Gen AI:** Machine Learning, NLP, LLMs, RAG, LangChain, LangGraph, Knowledge Graphs, TensorFlow, Pytorch, Keras, Scikit-learn, Vector Embeddings, Semantic Search, Azure Open AI
- **Data Engineering:** ETL Pipeline Development, Data Modeling, Query Optimisation, Lakehouse Architecture, Data Ingestion and Transformation
- **Databases:** PostgreSQL, MySQL, SQL Server, Azure SQL, Vector Databases
- **Cloud Platforms & DevOps:** AWS (Lambda, Glue, DynamoDB, RDS, S3, EC2, ECS, Aurora), Microsoft Fabric, Azure, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD
- **Analytics:** Semantic Models, Data Visualisation, Analytics Engineering, DAX, Data Reporting
- **Tools & Testing:** Pytest, TDD, API Testing, Performance monitoring, Git, GitHub, Postman

PROJECTS

Building a National Lottery Profit/Loss Simulation Platform

- Engineered a full-stack lottery simulation platform using **Python, Flask, SQLite, HTML, CSS, JavaScript, Bootstrap**, and **Plotly**, supporting **five** configurable lottery engines with dynamic probability and payout logics.
- Designed and implemented scalable **backend** workflows including secure authentication, session management, password hashing, and country-specific currency handling for **100+** global regions.
- Built interactive simulation analytics dashboards with **Plotly visualisation**, enabling real-time profit/loss tracking, spending analysis, and draw-by-draw financial reporting.
- Optimised performance through modular architecture, caching and efficient state management to deliver scalable multi-game simulation workflows.

NYSE Time Series Forecasting Model

- Engineered a multi-model time-series forecasting platform using **ARIMA, SARIMA**, and **LSTM** architectures to predict **NYSE** stock prices from large-scale historical market datasets.
- Designed and optimised end-to-end **ML** pipelines including data preprocessing, feature engineering, hyperparameter tuning, stationarity tuning, sequence modelling, and model evaluation (**RMSE, MAE, MSE**) using **Python, TensorFlow/Keras, Pandas** and **Scikit-learn**.
- Developed and deployed a Flask-based **web application** with real-time prediction capabilities, interactive **Plotly visualisations**, model integration, **caching**, and persistent deployment artifacts (.h5, .pkl) for scalable inference workflows.

Fruit Classification using TensorFlow and CNN

- Engineered a **computer vision** application using **Python, TensorFlow, Keras, CNNs, VGG16**, and **ResNet50** to classify fruit images across **131** categories from a dataset of **90,000+** images.
- Designed scalable **data pipelines** for image preprocessing, augmentation, dataset partitioning, and feature extraction, improving model robustness across diverse image conditions.
- Applied **transfer learning** and fine-tuning techniques on pre-trained **VGG16** and **ResNet50** architectures, significantly improving classification performance and reducing training overhead.
- Optimized model training through **hyperparameter tuning, RMSprop** optimization, dropout regularization, and categorical cross-entropy loss function for multi-class classification.
- Conducted comprehensive model evaluating using training/validation metrics and comparative analysis of **CNN, VGG16**, and **ResNet50** architectures to identify the best performing model.

EDUCATION

MSc – Data Science & AI (Merit)

Sep 2023 – Jul 2025

University of Liverpool

United Kingdom

- **Coursework:** Research Methods in Computer Science, Programming Fundamentals, Database and Information Systems, Web Programming, Data Mining and Visualisation, Maths and Statistics for AI and Data Science, Applied Artificial Intelligence.

BSc – Science (Distinction)

Jun 2018 – Sep 2021

Sri Venkateswara University

India

- **Coursework:** Computer Science, Mathematics and Physics.

CERTIFICATIONS

- Microsoft Certified: [SQL AI Developer Associate](#)
- Microsoft Certified: [Fabric Data Engineer Associate](#)
- Microsoft Certified: [Fabric Analytics Engineer Associate](#)
- Rest API (Intermediate) Certificate – HackerRank
- Problem Solving (Intermediate) Certificate – HackerRank